Nichole Yang

Work Rights: Australian Permanent Resident | Target Location: Melbourne & Sydney
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Summary

Senior Software Engineer with 6 years of experience designing enterprise backend systems, data platforms, and infrastructure automation at VMware and Citi. Contributed to a Claude-powered troubleshooting agent combining domain-aware RAG and read-only MCP diagnostics, improving Top-5 retrieval from 68% to 90%. Proven track record of delivering fault-tolerant architectures and scalable production systems while improving system resilience, data trustworthiness, and operational efficiency.

Tech Stack

Languages: Python, Java, C#, SQL, PowerShell, Shell, TypeScript
Frameworks & Platforms: Flask, Spring Boot, ASP.NET, Kubernetes, Docker, Linux, AWS(Certified Cloud Practitioner)

Work Experience

VMware - Senior Software Engineer *Shanghai, China | Oct 2021 - Jun 2026*
Designed and delivered distributed storage capabilities for vSAN across disaster recovery, telemetry analytics, platform reliability, AI-assisted troubleshooting, and customer-facing infrastructure automation.

Citi Group - Software Engineer *Shanghai, China | Aug 2020 - Oct 2021*
Developed Java backend services for an enterprise credit-risk analytics platform, implementing macroeconomic scenario functionality supporting regulatory stress testing, financial forecasting, and portfolio risk management.

Selected Projects

Agentic Troubleshooting and Failure intelligence Platform Python *Python | Java | Claude*

- Integrated a BERT-based failure classification and similarity-matching system into vSAN test workflows, automatically categorizing failed reports and clustering related historical cases to reduce manual triage across thousands of daily test reports.
- Extended the platform into a Claude-powered troubleshooting agent by designing domain-aware RAG to bridge symptom-to-root-cause terminology gaps and integrating read-only MCP diagnostics for live health and configuration; improved Top-5 recall from 68% to **90%**; deployed on VCF/VKS Kubernetes for **100+** team members and refined retrieval based on user feedback.

Enterprise Telemetry Platform *Python | SQL | Superset | Github Action & Jenkins*

- Led the architecture and evolution of a high-volume telemetry platform serving **55K+** deployments and **11K+** customers, owning end-to-end collection, Kafka-based ingestion, ETL pipelines, and analytics delivery powering **140+** business dashboards.
- Designed a runtime-configurable telemetry framework and re-architected high-volume data processing to improve collection scalability and release agility, reducing hourly memory peaks by **60%** without disrupting downstream analytics.
- Established automated CI/CD validation, end-to-end monitoring, and alerting, improving dashboard accuracy by 30%; coordinated producers and consumers and enabled **10+** engineers to independently build product dashboards and track feature adoptions.

Test Infrastructure Resource & Queue Management Platform *Python | Flask | Cursor*

- Designed and delivered a Flask-based resource orchestration platform integrated with internal infrastructure APIs using Cursor, supporting priority scheduling, quota, and automated resource lifecycle management; automated allocation and capacity monitoring, reducing infrastructure waste by 60% and improving operational visibility through usage tracking and bottleneck alerts.

Frontend & Full-Stack Projects *TypeScript | React*

- **vSAN Management UI**: Contributed to the disk management functionality for vSAN in vSphere Client management interface.
- **StockView** | Personal Project: Built a portfolio dashboard to address gaps in my own investing workflow, integrating read-only Schwab balances and positions, with custom holding groups, target tracking, and valuation snapshots.

Education

Carnegie Mellon University *Pittsburgh, United States | Aug 2018 - May 2020*
Master of Science in Information Technology - Business Intelligence and Data Analytics

Shanghai Lixin University of Accounting and Finance *Shanghai, China | Sep 2014 - Jun 2018*
Bachelor of Management in Information Management and Systems